

SINGO, Sweden

WMO: 024740

Lat: 60.167N

Lon: 18.750E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.6	-12.3	-17.5	0.8	-15.5	-14.4	1.1	-11.6	15.0	3.5	12.4	1.0	2.8	270	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	25.2	17.1	22.9	16.2	21.1	15.4	18.2	23.0	17.1	21.5	16.1	19.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	20.0	15.2	10.8	18.9	14.3	10.2	18.1	51.3	23.1	48.1	21.5	45.2	20.1	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.5	12.0	10.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-16.2	28.0	6.0	2.3	-20.5	29.7	-24.1	31.1	-27.5	32.4	-31.9	34.0		
			-16.5	19.6	5.9	1.5	-20.8	20.7	-24.3	21.6	-27.6	22.4	-31.9	23.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-1.8	-2.0	0.7	3.6	8.9	12.9	16.4	14.9	10.9	6.5	2.2	-0.6		
	DBStd	7.29	4.91	4.93	4.04	3.57	3.57	3.25	2.82	2.67	2.72	3.10	3.15	3.57		
	HDD10.0	1956	366	335	290	196	66	8	0	1	21	112	233	329		
	HDD18.3	4489	624	568	548	442	293	167	75	112	224	366	483	587		
	CDD10.0	533	0	0	0	4	32	94	198	152	48	4	0	0		
	CDD18.3	23	0	0	0	0	0	3	15	4	0	0	0	0		
	CDH23.3	158	0	0	0	0	5	31	100	21	1	0	0	0		
	CDH26.7	23	0	0	0	0	0	1	20	3	0	0	0	0		
(14) Wind	WSAvg	4.1	4.8	4.1	4.1	3.8	3.5	3.6	3.6	3.8	4.3	4.5	4.5	4.9		
(15) Precipitation	PrecAvg	572	38	35	29	34	38	55	60	71	52	56	55	50		
	PrecMax	751	68	73	63	78	73	106	157	176	130	117	114	100		
	PrecMin	436	4	7	0	4	9	12	14	4	12	15	11	10		
	PrecStd	75	19	15	16	22	17	27	39	38	28	27	26	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	9.2	11.7	19.5	23.8	26.2	29.3	25.9	21.3	14.0	10.4	7.9		
		MCWB	5.5	6.8	7.5	12.3	14.3	17.8	18.5	18.1	15.7	10.3	8.4	5.8		
	2%	DB	5.8	7.8	8.8	15.2	20.5	23.9	26.1	22.4	18.2	12.7	9.2	6.3		
		MCWB	3.4	5.7	5.3	9.2	13.1	15.8	17.5	16.9	13.3	9.9	7.2	4.6		
	5%	DB	4.7	5.7	7.0	11.8	18.3	21.3	23.6	20.5	16.5	11.8	7.7	5.2		
		MCWB	2.7	3.5	4.2	7.4	11.8	14.2	16.6	15.6	13.0	9.8	6.4	3.8		
	10%	DB	3.7	3.8	5.3	9.2	15.9	18.8	21.8	19.3	14.9	10.8	6.5	4.1		
		MCWB	2.1	2.4	3.0	6.2	10.5	13.4	15.9	14.8	11.7	9.1	5.2	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.4	8.3	12.4	15.2	19.4	19.7	18.9	16.1	11.4	8.8	5.9		
		MCDB	7.1	9.1	11.2	19.5	21.3	24.7	25.2	23.2	19.6	12.8	9.9	7.3		
	2%	WB	4.1	5.9	5.9	9.3	13.8	16.8	18.5	17.7	14.8	10.7	7.6	4.9		
		MCDB	5.3	7.5	7.9	14.0	19.9	22.0	23.8	21.6	17.7	12.0	8.7	6.2		
	5%	WB	3.1	3.9	4.3	7.9	12.0	15.1	17.5	16.6	13.4	10.0	6.4	4.0		
		MCDB	4.3	5.0	6.5	11.3	17.0	19.8	22.3	19.7	15.3	11.4	7.6	5.1		
	10%	WB	2.2	2.5	3.2	6.4	10.6	13.7	16.6	15.6	12.4	9.1	5.4	3.0		
		MCDB	3.3	3.5	5.1	9.1	15.3	18.4	21.0	18.2	14.3	10.6	6.4	4.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.0	5.2	6.5	8.6	7.7	7.7	6.3	6.0	4.6	3.7	4.3		
		MCDBR	4.8	5.8	7.2	11.2	13.1	11.9	10.8	8.2	7.9	5.9	4.7	4.6		
	5% WB	MCWBR	4.3	4.5	4.7	6.7	7.2	5.6	4.7	4.0	4.5	4.2	4.1	4.5		
		MCWBR	4.5	4.8	6.4	10.0	12.2	10.5	9.1	7.3	7.0	5.2	4.4	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.315	0.334	0.369	0.369	0.359	0.386	0.377	0.345	0.331	0.299	0.385		
	taud		2.203	2.283	2.347	2.340	2.409	2.475	2.421	2.441	2.515	2.421	2.236	2.662		
	Ebn at Noon		493	680	794	828	858	873	836	815	785	661	453	350		
	Edh at Noon		41	67	86	103	104	99	102	93	73	59	40	29		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.80	2.31	3.97	5.37	6.01	5.60	4.36	2.72	1.17	0.34	0.16		
	RadStd		0.03	0.15	0.25	0.35	0.43	0.38	0.54	0.37	0.28	0.18	0.08	0.02		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-110	-130	N/A	N/A	

Nomenclature: See separate page